MOHAN BABU UNIVERSITY

Anveshana Club

Explore · Build · Bring Ideas to Life.

PRODUCT REQUIREMENTS DOCUMENT (PRD)

VERSION: 1.0
DATE: [TBD]
STATUS: FINAL DRAFT

1. Document Purpose, Version, and Scope

Purpose: This Product Requirements Document (PRD) serves as the comprehensive source of truth for the design, development, and launch of the official Anveshana Club website at Mohan Babu University (MBU), Tirupati. It defines the core features, constraints, and business logic required for a successful minimum viable product (MVP).

Version: 1.0 (Initial Public Release)

Scope: The scope includes a responsive, public-facing web portal for event discovery, club visibility, team introductions, and an administrative dashboard for content management. Custom internal registration systems, payment gateways, and student social networks are explicitly out of scope.

2. Product Overview and Vision

Anveshana is a central university club uniting students from the School of Computing and the School of Engineering. Rooted in the tagline *"Explore · Build · Bring Ideas to Life,"* the club fosters innovation and cross-disciplinary collaboration.

Vision: To create a premium, visually striking digital presence that acts as the front door to the Anveshana Club. The website will leverage a cinematic, editorial aesthetic (navy and golden-orange) to attract students, showcase high-quality event photography, and streamline event participation through frictionless, no-login discovery.

3. Problem Statement and Proposed Solution

The Problem: Currently, Anveshana relies on fragmented communication channels (WhatsApp groups, physical posters) to announce events and showcase past achievements. This results in poor club visibility among new students, difficult event tracking, and a lack of a centralized, professional repository for club activities and leadership profiles.

The Solution: A centralized, publicly accessible web application requiring zero student friction (no accounts/login). It will serve as an elegant informational hub featuring event schedules, photo galleries, direct external registration links (Google Forms), and an administrative backend for seamless content updates.

4. Target Users and User Personas

The product serves two distinct user groups with entirely different access paradigms:

> **Persona 1: The Curious Student (Primary, Public)**
> *"I want to know what this club does and how I can participate in their upcoming hackathon."*
> Needs: Immediate access to event details, visual proof of past success (galleries), clear instructions on how to register, and ability to submit feedback. Does not want to create yet another university account.

> **Persona 2: The Club Admin / Core Team (Secondary, Authenticated)**
> *"I need to publish the new event banner and update the Google Form link before our marketing push at 5 PM."*
> Needs: A hidden, secure `/admin` route. Easy CRUD (Create, Read, Update, Delete) capabilities for events, team member profiles, and homepage content without requiring code deployments.

5. Product Goals and Measurable Success Metrics

- **Goal 1: Maximize Discoverability.** Achieve 1,000+ unique student visits within the first month of the academic semester.
- **Goal 2: Frictionless Event Registration.** Maintain a click-through rate (CTR) of >30% from Event Detail pages to the Google Form registration links.
- **Goal 3: Streamlined Administration.** Reduce the time required to publish a new event from hours (via manual campus outreach) to under 5 minutes via the admin dashboard.
- **Goal 4: Brand Authority.** Establish a premium digital footprint that aligns with MBU's standards, measured by zero negative accessibility or mobile-responsiveness feedback.

6. Functional Requirements (FR)

The following table outlines the core functionalities required for the system. Each requirement is mandatory for the MVP unless noted otherwise.

Req ID	Feature Area	Description
FR-001	Public Access	The entire public-facing website SHALL be accessible without requiring user authentication, login, or account creation.
FR-002	Home Page	The system SHALL display the club's mission, tagline, upcoming event highlights, recent completed events, and global navigation.
FR-003	Event Catalog	The system SHALL display an index of events categorized by status (Upcoming, Ongoing, Completed) organized chronologically.
FR-004	Event Details	The system SHALL generate a dedicated page per event displaying banner, description, venue, date, eligibility, and status.
FR-005	Registration Link	The system SHALL provide a direct outward link (e.g., Google Forms) for event registration when the event status is "Open".
FR-006	Galleries	The system SHALL support dedicated image galleries linked to specific completed events.
FR-007	Core Team Directory	The system SHALL display team members organized into 7 expandable categories (Leads, Infra, Design, Data, Social, PR, Tech) with profile links.
FR-008	Feedback System	The system SHALL provide a public form for 1-5 star ratings and text feedback, secured against basic spam.
FR-009	Admin Routing	The system SHALL obscure administrative access behind a specific route (`/admin`) not linked in the public UI.
FR-010	Content Management	The admin dashboard SHALL allow authenticated users to perform CRUD operations on Events, Galleries, Team Members, and Feedback.

7. Non-Functional Requirements (NFR)

Req ID	Category	Description
NFR-001	Performance	The website SHALL load its Largest Contentful Paint (LCP) in under 2.5 seconds on a standard 4G mobile connection. Image assets must be optimized/compressed.
NFR-002	Responsiveness	The UI SHALL be fully responsive, prioritizing a mobile-first layout suitable for students browsing on smartphones, while scaling elegantly to desktop monitors.
NFR-003	Security	The `/admin` route SHALL be protected by secure authentication (e.g., JWT or robust session management). Passwords MUST be hashed (e.g., bcrypt). Form endpoints MUST have CSRF/spam protection.
NFR-004	Aesthetic	The UI SHALL strictly adhere to the defined palette: Navy and Golden-Orange, utilizing premium typography (serif headers, clean sans-serif body) to mirror a cinematic, editorial style.
NFR-005	Availability	The website architecture SHALL support standard shared or cloud hosting environments (e.g., Vercel, Netlify, or standard VPS) targeting 99.9% uptime.

8. User Stories

Public Users (Students)

- **US-01:** As a student, I want to see the club's mission and latest events on the homepage, so that I can quickly understand what Anveshana is about.
- **US-02:** As a student, I want to filter events by "Upcoming" or "Completed", so that I can find activities I can join versus viewing past successes.
- **US-03:** As a student, I want to click a "Register" button on an event page that takes me directly to a Google Form, so that I can sign up without creating a new website account.
- **US-04:** As a student, I want to view high-quality photos from past events, so that I can see the club's culture and scale.
- **US-05:** As a student, I want to view the core team organized by department, so that I know who to contact for specific queries (e.g., PR or Tech).
- **US-06:** As a student, I want to leave a star rating and anonymous feedback, so that I can share my thoughts on a recent hackathon.

Administrative Users (Core Team)

- **US-07:** As an admin, I want to log in through a hidden `/admin` URL, so that the public site remains clean and focused on students.
- **US-08:** As an admin, I want to add a new event with a banner, description, and external registration link, so that students can see it immediately without requiring a developer to update code.
- **US-09:** As an admin, I want to upload a batch of photos to a specific event gallery, so that the event record is visually rich.
- **US-10:** As an admin, I want to update team member profiles (e.g., changing a GitHub link or designation), so that the team directory is always current.
- **US-11:** As an admin, I want to read submitted feedback and ratings in a dashboard, so that we can improve future club activities.

9. Feature Prioritization (MoSCoW)

Must Have (MVP)	Should Have
• Public homepage with branding. • Event listing & detail pages. • External registration links (Google Forms). • Core team directory (7 categories). • Hidden Admin dashboard for basic CRUD.	• Event photo galleries. • Feedback and 5-star rating form. • Rich text editing for event descriptions. • Spam protection on feedback forms.
Could Have (Future)	**Won't Have (Out of Scope)**
• Dynamic filtering by event tags. • Automated email newsletters. • Analytics dashboard integrated into `/admin`.	• Internal student accounts / logins. • Payment gateways / ticketing. • Internal chat or forums. • Native internal registration system.

10. Detailed Page and Feature Requirements

10.1 Home Page

The visual anchor of the site. Must feature a cinematic hero banner (campus sunset aesthetic). Must include:

- **Hero Section:** Oversized serif Anveshana title, tagline ("Explore · Build · Bring Ideas to Life").
- **About Snippet:** Brief mission statement bridging SoC and SoE.
- **Dynamic Sections:** "Upcoming Events" (shows next 3 chronologically), "Recent Highlights" (shows last 3 completed).
- **Global Footer:** Official contact info (TBD placeholders), social links.

10.2 Events & Galleries

Must support distinct states: Upcoming, Open for Registration, Closed, Completed. The UI will render 'Event Cards' displaying the banner, title, and a status badge. Detailed pages must accommodate a large hero image, structured metadata (Date, Time, Venue, Eligibility), and a prominent CTA button mapping to the Google Form URL.

10.3 Core Team Directory

A structured accordion or tabbed interface enforcing exactly 7 categories: *Team Leads, Infrastructure, Design, Data Management, Social Media, PR, Technical*. Profile cards must standardize inputs: Photo, Name, Designation, Bio (max 150 chars), and icon-links for LinkedIn, GitHub, and Portfolio.

11. Admin Permissions & Content Management

The backend empowers non-technical core team members to manage the site. Security and usability are paramount.

- **Access Control:** Navigating to `/admin` prompts a login screen. No links to this route will exist in the public DOM or sitemap.
- **Authentication:** Requires a valid admin username/email and password. For MVP, a single shared admin account or predefined role-based accounts (Admin vs. Editor) are acceptable, depending on backend technical selection.
- **Event Management:**
  - Create/Edit form including fields for: Title, Description, Date, Time, Venue, Eligibility, Image Upload (Banner), Registration URL, Status Dropdown.
  - Toggle visibility (Draft vs. Published).
- **Gallery Management:** Associate a batch of uploaded image URLs to a specific 'Completed' event.
- **Team Management:** Add/Remove members, assign them to one of the 7 fixed categories, and set display order (drag-and-drop or priority integer).
- **Feedback Review:** A read-only table displaying submitted ratings, text feedback, and timestamps.

12. Business Rules and Validation

12.1 Event Registration Logic

The system relies on Google Forms for registration. Therefore, the website's responsibility is purely routing and state display:

- If Event Status = `Upcoming` (no link yet), button reads "Registrations Opening Soon" (Disabled).
- If Event Status = `Open`, button reads "Register Now" (Active) and links to the provided URL via `target="_blank"`.
- If Event Status = `Closed` or `Completed`, button reads "Registrations Closed" (Disabled).
- **Constraint:** The system SHALL NOT redirect users to WhatsApp groups for primary event registration.

12.2 Form Validation (Feedback)

- Rating (1-5 stars) is REQUIRED.
- Feedback Text is OPTIONAL, capped at 500 characters.
- Submissions must be rate-limited (e.g., max 3 submissions per IP per hour) to prevent database spamming.

12.3 Data Integrity

- Images uploaded via admin must be validated for type (JPEG, PNG, WebP) and size (max 2MB per image) before saving to storage.
- Unknown or undefined official details (e.g., club email, phone number) must render dynamically or default to "TBD" on the frontend; do not hardcode mock data in production.

13. Accessibility, Responsiveness, and Security

13.1 Accessibility (a11y)

The platform must comply with baseline WCAG 2.1 AA standards. Ensure sufficient color contrast between the navy (`#0A192F`) background and golden-orange (`#D4AF37`) or white text. All images (event banners, team photos) must have descriptive `alt` attributes. Interactive elements must be fully keyboard navigable.

13.2 Responsiveness

The site must employ a responsive grid architecture. Given the target demographic (university students), mobile traffic is expected to exceed 80%. Multi-column layouts on desktop (e.g., event grids, team members) must stack gracefully to single columns on devices under 768px width.

13.3 Security Expectations

Although the public site handles minimal PII (only optional feedback), the admin dashboard is a critical attack vector. Implement HTTPS everywhere. Admin endpoints must enforce strict authorization headers. Database inputs must be sanitized to prevent SQL/NoSQL injection and Cross-Site Scripting (XSS) via the rich-text event descriptions.

14. Assumptions, Constraints, and Risks

- **Assumption:** Google Forms will remain the standard for data collection, avoiding the need for complex internal GDPR/data-privacy compliance regarding student PII.
- **Assumption:** The university administration approves of the club managing its own independent web presence outside the primary MBU IT infrastructure.
- **Constraint:** The project must be maintainable by future cohorts of students; hence, the architecture should favor popular, well-documented frameworks (e.g., React, Next.js, Node.js) over niche technologies.
- **Risk:** Admin credentials being shared and leaked among students. *Mitigation:* Implement strong password policies and ideally session timeouts.
- **Risk:** High storage costs from event photo galleries. *Mitigation:* Enforce aggressive client-side/server-side image compression before cloud storage (e.g., Cloudinary/AWS S3).

15. Explicit Out-of-Scope Features

To ensure successful MVP delivery, the following features will NOT be built in this version:

- Student user accounts, profiles, or login portals.
- Internal payment processing or ticketing systems.
- In-app messaging, forums, or internal chat systems.
- Direct hosting of video content (must use external embeds like YouTube if required).
- Native event registration routing (all registration data lives purely on Google Forms).

16. Acceptance Criteria

The project is deemed complete when the following criteria are met and verified by the Product Manager / Core Team Leads:

> **AC1: Public Navigation & Branding**
> Given a user visits the homepage, the cinematic navy/gold theme is applied, the Anveshana logo and tagline are visible, and no login prompt interrupts the experience.

> **AC2: Event Discovery & Registration**
> Given a user clicks on an "Open" event, they are taken to a detailed view, can read all metadata, and clicking "Register" successfully opens a new tab with the designated Google Form.

> **AC3: Admin Obfuscation & Security**
> Given a user inspects the site menu or footer, no link to the admin panel exists. Navigating manually to `/admin` requires a valid authentication challenge.

> **AC4: Content Management (No-Code Updates)**
> Given an authenticated admin uses the dashboard, they can successfully create a new event, assign it to a category, and verify it appears on the public homepage immediately upon publishing.

> **AC5: Team Hierarchy**
> Given a user navigates to the Team section, they see exactly 7 predefined departments, expandable/collapsible, populated with member cards.

17. Launch-Readiness Checklist

- **Design & UX:** Final UI review against the AICTE Portal benchmark and internal cinematic mood board. All placeholder copy replaced or marked explicitly as TBD.
- **Technical:** Mobile responsiveness tested across iOS/Android devices. Cross-browser compatibility confirmed (Chrome, Safari, Firefox).
- **Performance:** Lighthouse performance score >85. Image compression pipeline active.
- **Security:** `/admin` route tested for unauthorized access bypass. Form endpoints tested for basic spam/injection resilience.
- **Content:** Initial dataset seeded (at least 1 upcoming event, 2 past events with galleries, full core team directory populated).
- **Infrastructure:** Domain configured (e.g., anveshana-mbu.in or subdomain). SSL certificate provisioned and active.

"Explore · Build · Bring Ideas to Life."